

The Capstone Project: From Specialist to Professional

Integrating Technology, Process, and Teamwork

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Course Goal

- To simulate the modern software development lifecycle
- To help you grow from specialists to professionals
  - within the comfort zone of a sandbox
- Why This Matters (Especially for You)



SE



Security



A vulnerability found and fixed after deployment is a failure
A vulnerability prevented by a robust process is a professional success

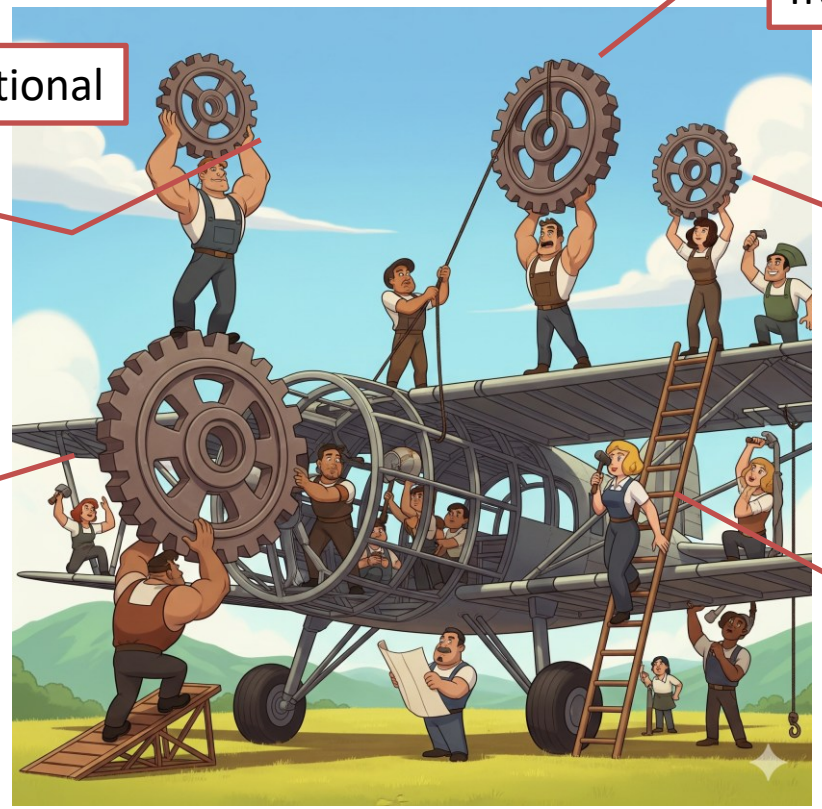
Core Learning Objectives



Master Professional Teamwork

Execute Roles in a Development Method

Integrate Your Technical Expertise



functional

security

non-functional

performance

TESTING

Node.js

frameworks

React

advanced Git workflows

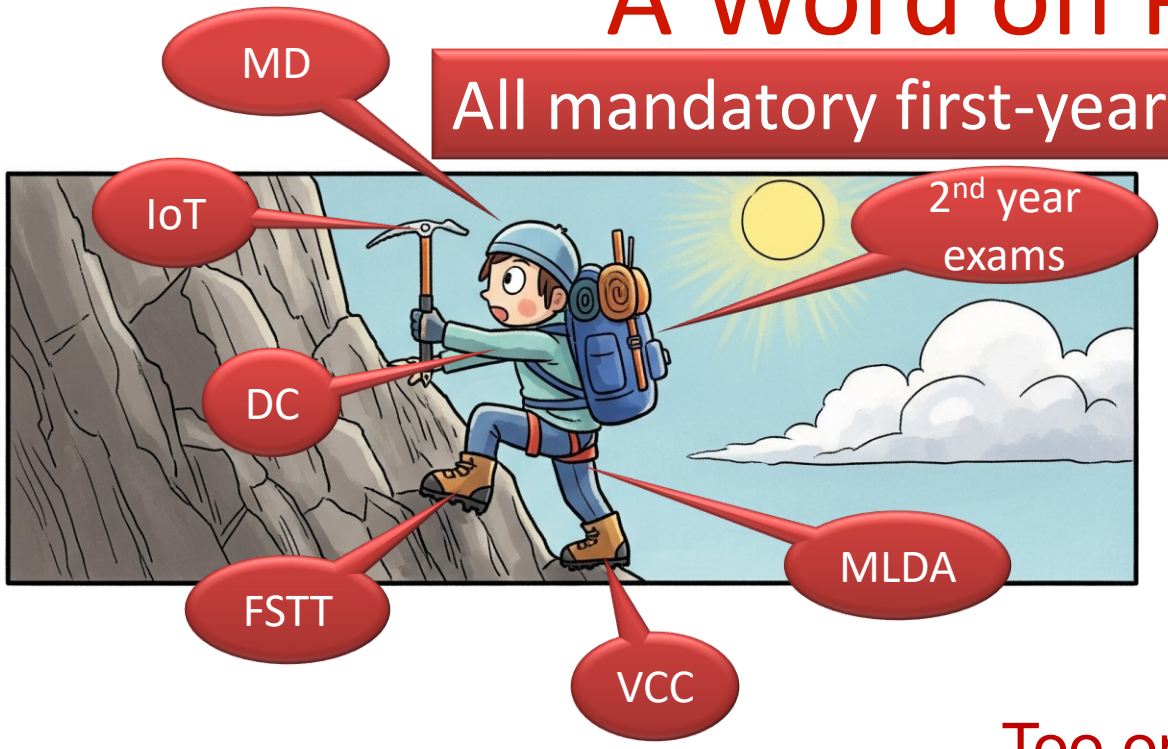
Docker

containerization

patterns

A Word on Prerequisites

All mandatory first-year exams must be passed





Without prerequisites

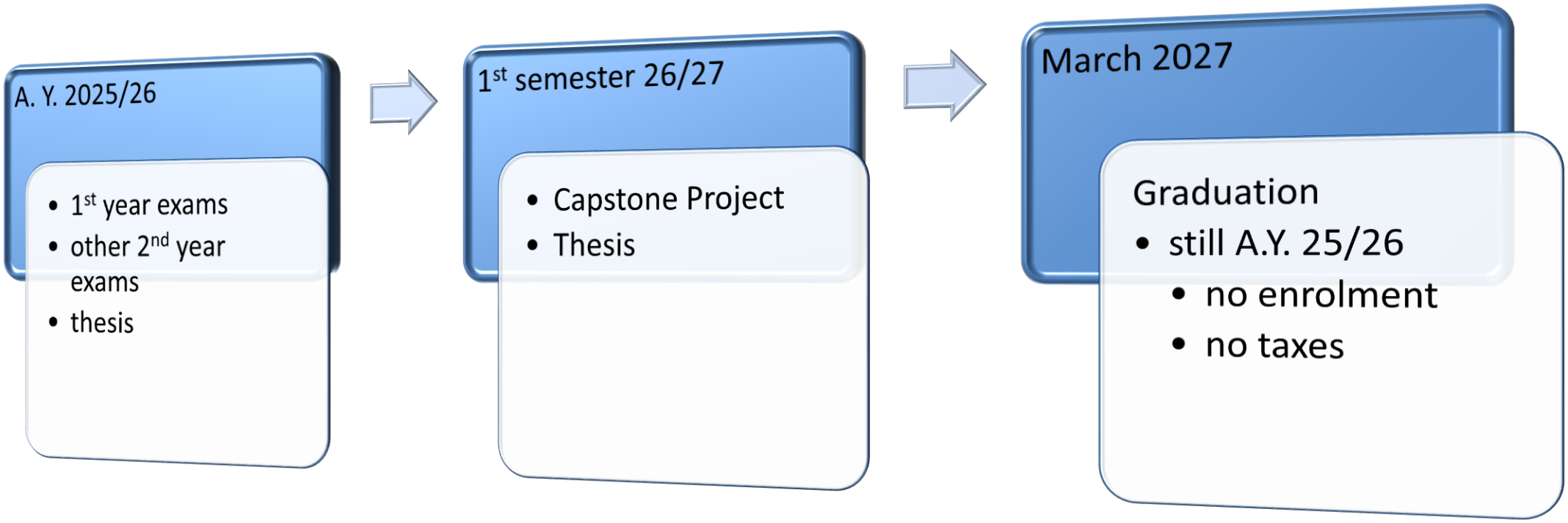
Too optimistic



Without prerequisites

Postponing Capstone Project

Not as dramatic as you could think



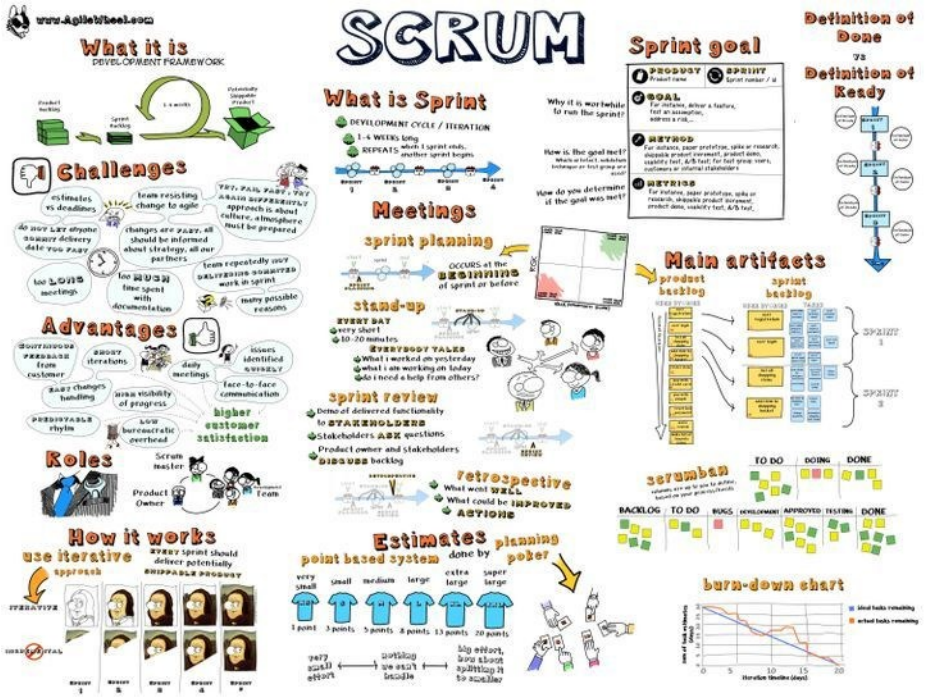
How Teams Will Be Formed



- Goal: effective, balanced, and homogeneous teams
  - no isolation of outliers
  - shared effort $\Rightarrow$ shared growth
- Classification based on
  - marks in relevant prerequisite courses
  - your participation into the activities of the initial period

Course Structure & Topics

- Focus is on the **PROCESS**
- Main Topics
  - User Stories
  - Scrum
  - User Interaction (UI/UX)
  - User Testing

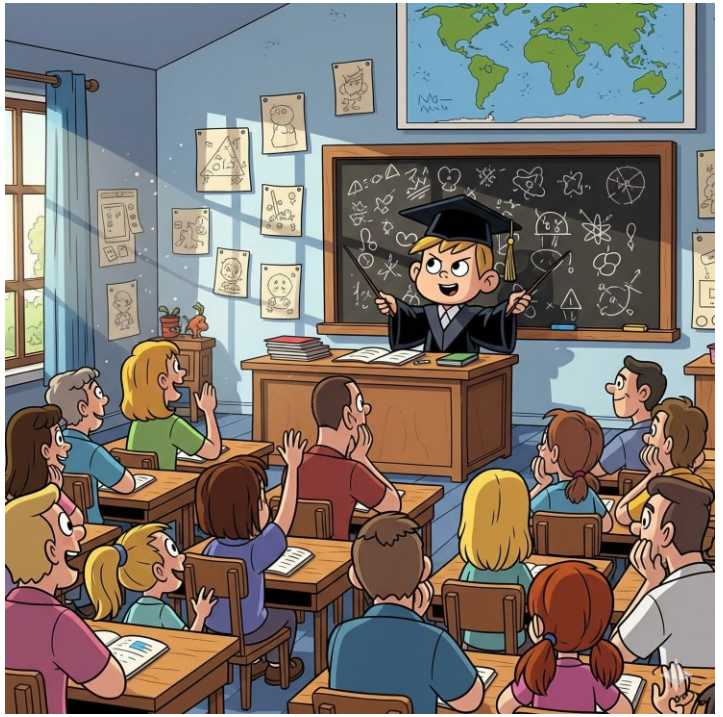




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A *Flipped Classroom* Approach

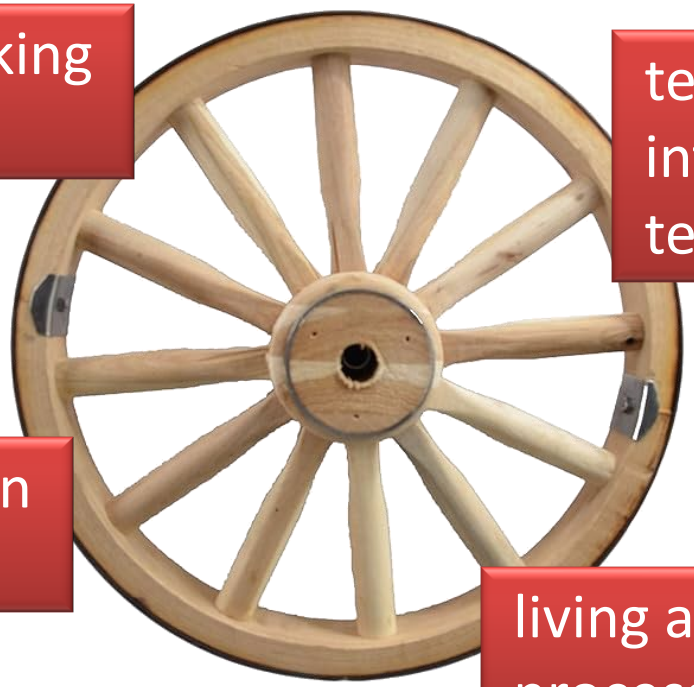
- For each topic
  - Brief Intro
    - I give a short introductory lecture
  - Self-Study
    - you complete a mandatory list of readings and videos
  - Student-Led Discussion
    - student presentation(s) on advanced aspects of the topic
    - plenary discussion
  - Hands-On Exercises
    - you apply the concepts in practical scenarios
  - Topic Exam
    - a short, graded exam



The Project: Your Knowledge Hub

Where theory becomes reality

realistic team-working experience

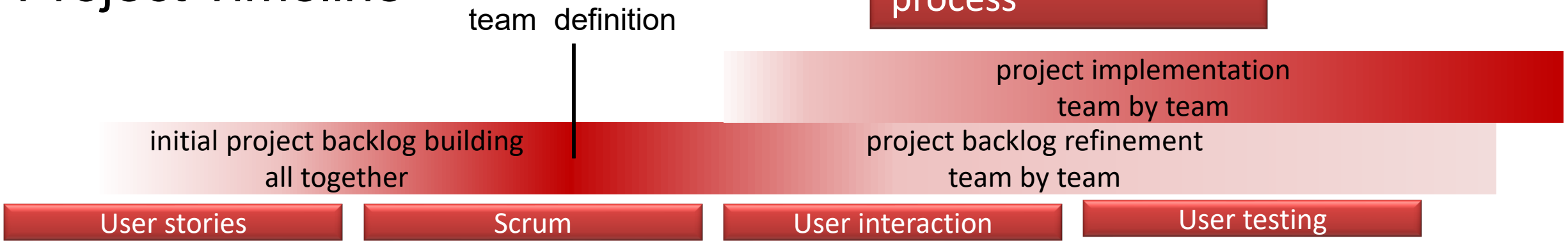


test-bed for your integrated technological skills

extensive application of all course topics

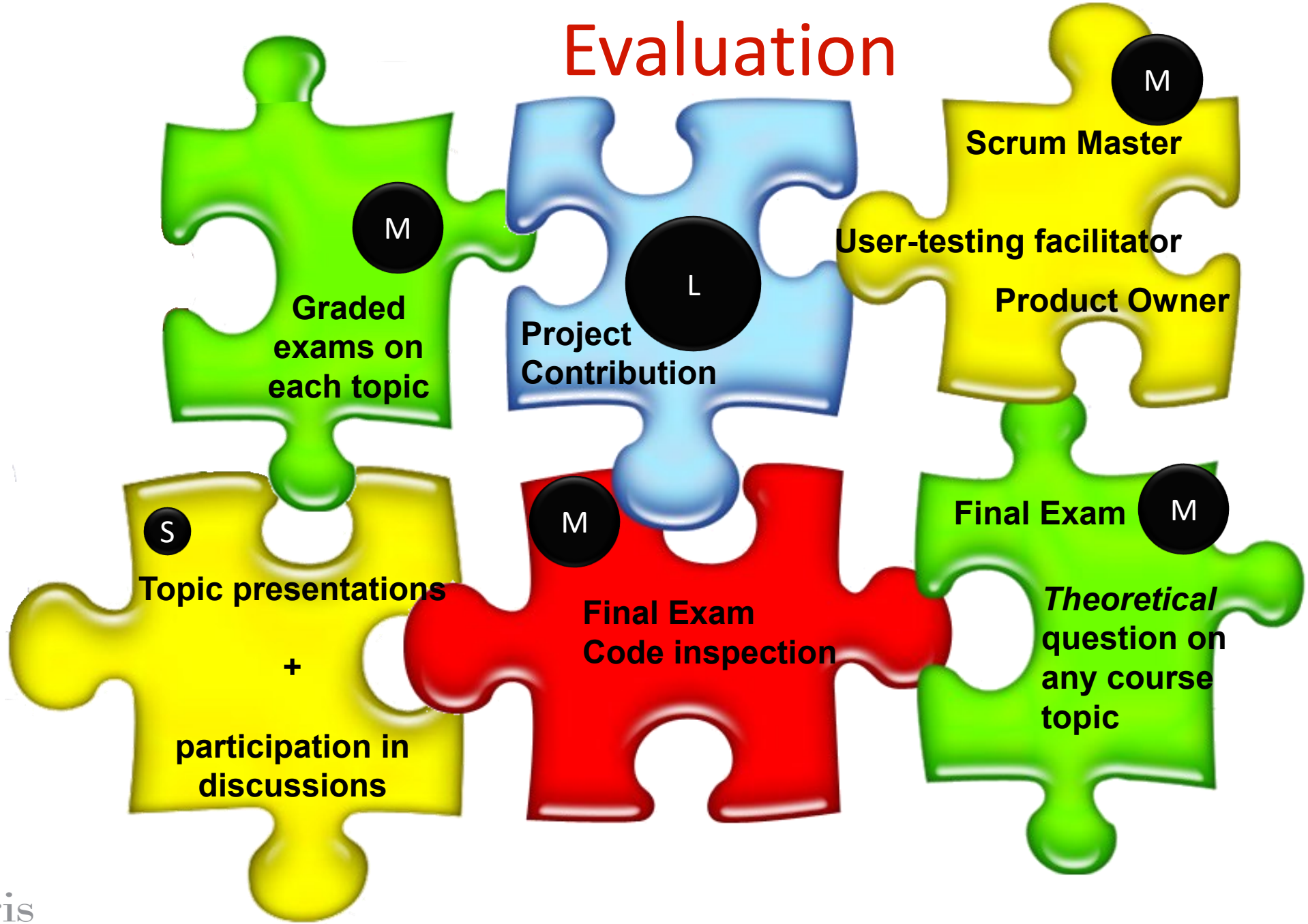
living a development process

Project Timeline



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Evaluation


M
Scrum Master
User-testing facilitator
Product Owner
M
Graded exams on each topic
L
Project Contribution
S
Topic presentations
+
participation in discussions
M
Final Exam
Code inspection
Final Exam
M
Theoretical question on any course topic

I Don’t Like my Mark

- Marks are not negotiable
- Missed topic exams can be replaced by extra questions during the final exam
- The exam can be taken again but
  1. *replacement*, not second chance
  2. you cannot make a better second *first impression*
     - only the *theoretical question* can be done again

     ⇒ significant mark improvements require taking the course again next year

Next Steps & Q&A

- Next Step: Our first assessment activity will be discussed tomorrow
- Check the course website for
  - the flipped lesson schedule
  - the first set of mandatory readings
- Questions?

After tomorrow's lesson



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